

SOFTWARE REQUIREMENTS

Chat Application

Version 1.0 | May 2026

1. Introduction

1.1 Purpose

This Software Requirements Specification (SRS) describes the functional and non-functional requirements for a mobile/web Chat Application. It is intended to guide the design, development, and testing of the system.

1.2 Scope

The Chat Application will allow registered users to:

- Securely create accounts and log in
- View and search their list of conversations
- Send and receive real-time messages
- Manage their personal profile

1.3 Definitions & Acronyms

Term	Definition
SRS	Software Requirements Specification
UI	User Interface
Auth	Authentication / Authorization
FR	Functional Requirement
NFR	Non-Functional Requirement

1.4 Overview

Section 2 describes the overall system; Section 3 details functional requirements per screen; Section 4 covers non-functional requirements; Section 5 lists constraints; Section 6 shows the navigation flow.

2. Overall System Description

2.1 System Perspective

The Chat Application is a standalone client-server system. The client (mobile or web) communicates with a backend server over secure HTTPS/WebSocket connections. Data is persisted in a cloud database.

2.2 User Classes

User Class	Description
Guest	Unauthenticated visitor; can access Login and Signup screens only.
Registered User	Authenticated user with full access to Chat List, Individual Chat, and Profile screens.

2.3 Assumptions & Dependencies

- Users have access to a modern mobile device or browser with internet connectivity.
- The backend server supports REST APIs and WebSocket connections.
- A third-party email service may be used for account verification.

3. Functional Requirements

3.1 Screen 1: Login Screen

The Login Screen is the entry point of the application for returning users.

ID	Requirement	Description	Priority
FR-1.1	Email Field	The system shall provide a text input field for the user to enter their registered email address.	High
FR-1.2	Password Field	The system shall provide a masked password input field.	High
FR-1.3	Login Button	The system shall validate credentials and navigate to the Chat List Screen on success, or display a message on failure.	High
FR-1.4	Signup Navigation	The system shall display a 'Don't have an account?' link that navigates to the Signup Screen.	High
FR-1.5	Input Validation	The system shall validate email format and ensure the password field is not empty before submission.	Medium

3.2 Screen 2: Signup Screen

The Signup Screen allows new users to create an account.

ID	Requirement	Description	Priority
FR-2.1	Name Field	The system shall provide a text input field for the user's full name.	High
FR-2.2	Email Field	The system shall provide an input field for the user's email address.	High
FR-2.3	Password Field	The system shall provide a masked password creation field.	High
FR-2.4	Confirm Password	The system shall provide a confirm password field and validate that both entries match.	High
FR-2.5	Register Button	The system shall create a new user account on successful validation and redirect to the Login Screen.	High

ID	Requirement	Description	Priority
FR-2.6	Duplicate Check	The system shall prevent registration with an already-registered email address.	Medium

3.3 Screen 3: Chat List Screen

The Chat List Screen displays all conversations of the authenticated user.

ID	Requirement	Description	Priority
FR-3.1	Search Bar	The system shall provide a search bar to filter conversations by user name or message content.	High
FR-3.2	Profile Picture	Each conversation entry shall display the contact's profile picture as a thumbnail.	Medium
FR-3.3	Last Message	Each conversation entry shall display a preview of the most recent message.	High
FR-3.4	Timestamp	Each conversation entry shall display the time or date of the last message.	Medium
FR-3.5	Unread Badge	The system shall display an unread message count badge for conversations with unread messages.	High
FR-3.6	Navigation	Tapping a conversation entry shall navigate the user to the Individual Chat Screen.	High

3.4 Screen 4: Individual Chat Screen

The Individual Chat Screen facilitates real-time messaging between two users.

ID	Requirement	Description	Priority
FR-4.1	Header Info	The screen shall display the contact's name, profile picture, and online/offline status.	High
FR-4.2	Message History	The system shall display the full message history in chronological order with sent/received messages.	High
FR-4.3	Message Input	The system shall provide a text input box for composing new messages.	High
FR-4.4	Send Button	The system shall provide a Send button that transmits the composed message to the recipient in real time.	High
FR-4.5	Emoji Button	The system shall provide an emoji picker button that opens an emoji selection panel.	Medium
FR-4.6	Attachment Option	The system shall provide an attachment option allowing users to send files, images, or media.	Medium
FR-4.7	Real-time Delivery	Messages shall be delivered and displayed in real time using WebSocket or equivalent protocol.	High

3.5 Screen 5: Profile Screen

The Profile Screen allows users to view and manage their personal account information.

ID	Requirement	Description	Priority
FR-5.1	Profile Photo	The system shall display the user's current profile photo and allow them to update it.	Medium
FR-5.2	Name & Email	The system shall display the user's name and registered email address.	High
FR-5.3	Edit Profile	The system shall allow the user to edit their name and profile photo; updated data shall be saved to the database.	High

ID	Requirement	Description	Priority
FR-5.4	Logout	The system shall provide a Logout button that terminates the session and redirects to the Login Screen.	High

4. Non-Functional Requirements

4.1 Performance

- Messages shall be delivered within 500 ms under normal network conditions.
- The Chat List Screen shall load within 2 seconds for up to 200 conversations.
- The application shall support a minimum of 1,000 concurrent users without degradation.

4.2 Security

- All data transmitted between client and server shall be encrypted using TLS 1.2 or higher.
- Passwords shall be stored using a strong hashing algorithm (e.g., bcrypt with salt).
- User sessions shall expire after 30 minutes of inactivity.
- The system shall implement rate limiting on authentication endpoints to prevent brute-force attacks.

4.3 Usability

- The UI shall be responsive and functional on screens from 4 inches to desktop size.
- All interactive elements shall have a minimum tap target size of 44x44 points.
- Error messages shall be clear, specific, and actionable.

4.4 Reliability

- The system shall achieve 99.5% uptime (excluding scheduled maintenance).
- The application shall gracefully handle loss of network connectivity and reconnect automatically.

4.5 Scalability

- The backend architecture shall be horizontally scalable to accommodate user growth.
- The database shall support efficient querying of message history for up to 10,000 messages per conversation.

5. Constraints & Limitations

- The MVP (v1.0) supports one-to-one (direct) messaging only; group chats are out of scope.
- Attachment file size shall be limited to 10 MB per file.
- The application will support English language only for v1.0.
- Supported platforms: iOS 14+, Android 10+, and modern web browsers (Chrome, Firefox, Safari, Edge).

6. Screen Navigation Flow

The following table describes the navigation model derived from the wireframe design:

From Screen	To Screen	Trigger
Login Screen	Chat List Screen	Successful login
Login Screen	Signup Screen	Tap 'Don't have an account?'
Signup Screen	Login Screen	Successful registration
Chat List Screen	Individual Chat Screen	Tap a conversation entry
Chat List Screen	Profile Screen	Tap profile / settings icon
Profile Screen	Login Screen	Tap Logout button

7. Revision History

Version	Date	Author	Description
1.0	May 2026	Development Team	Initial draft based on wireframe design